

46 YEARS OF EXCELLENCE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-14** **Date: 08.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

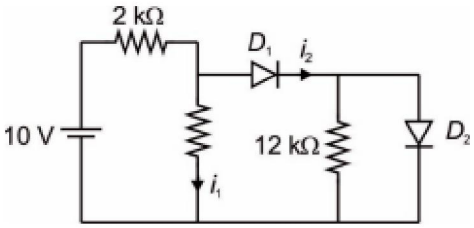
01. A parallel beam of uniform monochromatic light of wavelength 2640 $\overset{0}{A}$ has an intensity of $100 Wm^{-2}$. The number of photons in $1 mm^3$ of this radiation is(approximately)

A) 222 B) 335 C) 442 D) 555

02. In the following circuit if D_1 and D_2 are ideal diodes, then the values of currents i_1 and i_2 are respectively (consider the unknown resistance to have any non-zero resistance):

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A) $0,0$ B) $5mA,5mA$ C) $5mA,0$ D) $0,5mA$

03. Three quantities A, B and C are measured as $A=2\pm0.005, B=1\pm0.001$ and $C=4\pm0.01$. Then the percentage error in $R=\dfrac{AC^3}{B^2}$ is

A) 1.2 B) 0.8 C) 0.7 D) 1.05

04. An inductance L, capacitor C and a resistor R may be connected to an AC source of angular frequency ω in three different combinations of RC, RL and LC in series. Assume that $\omega L=\dfrac{1}{\omega C}$. The power drawn by the three combinations are P_1, P_2, P_3 respectively. Then

A) $P_1>P_2>P_3$ B) $P_1=P_2<P_3$ C) $P_1=P_2>P_3$ D) $P_1=P_2=P_3$

05. The volume of a closed vessel A is twice the volume of another closed vessel B, and both of them are filled with the same gas. If the gas A is at twice the temperature and twice the pressure in comparison to the gas in B, then the ratio of the number gas molecules in A and B is

A) 1 : 4 B) 4 : 1 C) 2 : 1 D) 1 : 2

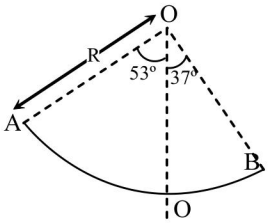
06. Equation of SHM is $x=10\sin10\pi t$. Find the distance between the two points where speed is $50\pi\ cm/s$. x is in cm and t is in seconds.

A) 10 cm B) 14 cm C) 17.32 cm D) 8.66 cm

07. A section of fixed smooth circular track of radius R in vertical plane is shown in the figure. A block is released from position A and leaves the track at B. The radius of curvature of its trajectory just after leaving the track at B is-

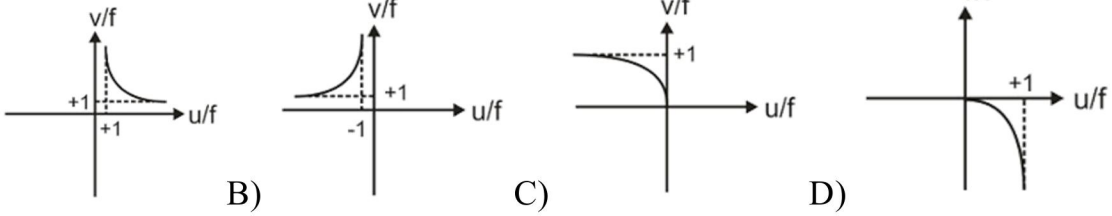
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A) R B) R/4 C) R/2 D) none

08. A real inverted image in a concave mirror is represented by (u, v, f are coordinates of object, image and the focus respectively with sign convention):

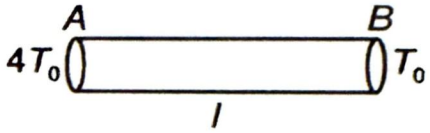


A) B) C) D)

09. A proton, a deuteron and an α -particle with the same KE enter a region of uniform magnetic field moving at right angles to B. What is the ratio of the radii of their circular paths –

A) $1:\sqrt{2}:1$ B) $1:\sqrt{2}:\sqrt{2}$ C) $\sqrt{2}:1:1$ D) $\sqrt{2}:\sqrt{2}:1$

10. The two ends A and B of a rod of length l and of uniform cross-sectional area are kept at temperature $4T_0$ and T_0 respectively. The thermal conductivity of the rod varies as $k = aT$, where a is a constant and T is the absolute temperature. The temperature at the centre of the rod during steady state is:



A) $\frac{5T_0}{2}$ B) $\sqrt{\frac{13}{2}}T_0$ C) $\sqrt{\frac{15}{2}}T_0$ D) $\sqrt{\frac{17}{2}}T_0$

11. A large number of liquid drops of radius 'a' coalesce to form a single spherical drop of radius b. The energy released in the process is converted into the kinetic energy of the

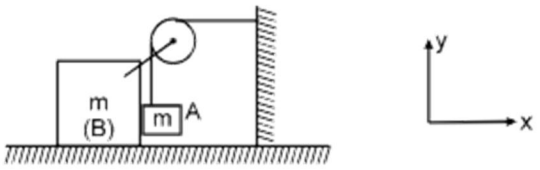
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big drop formed. If ρ and T are the density and surface tension of liquid respectively then the speed of big drop is-

A) $\sqrt{\frac{6T}{\rho}\frac{(b-a)}{ab}}$ B) $\sqrt{\frac{4T}{\rho}\frac{(b-a)}{ab}}$ C) $\sqrt{\frac{2T}{\rho}\frac{(b-a)}{ab}}$ D) $\sqrt{\frac{3T}{\rho}\frac{(b-a)}{ab}}$

12. In the system shown block A is released from rest. The acceleration of block A once released is



A) $\frac{g}{2}\hat{i}-\frac{g}{2}\hat{j}$ B) $\frac{g}{3}\hat{i}$ C) $\frac{g}{3}\hat{i}-\frac{g}{3}\hat{j}$ D) $\frac{g}{2}\hat{i}$

13. A wire elongates by x mm when a load W is hanged from it. If the wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be (in mm) –

A) 0 B) $x/2$ C) x D) 2 x

14. A geo-stationary satellite is orbiting the earth at a height of 6 R above the surface of the earth, R being the radius of the earth. The time-period of another satellite at a height of 2.5 R from the surface of earth is –

A) $6\sqrt{2}$hr B) 6 hr C) $5\sqrt{2}$hr D) 10 hr

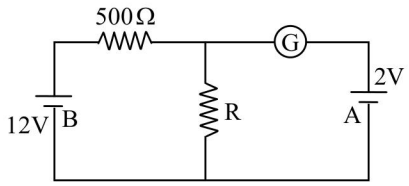
15. What will be the height of the image when an object of 2mm is placed at a distance 20 cm in front of the axis of a convex mirror of radius of curvature 40 cm?

A) 1.5 mm B) 2 mm C) 3 mm D) 1 mm

16. In the circuit, the galvanometer G shows zero deflection. If the batteries A and B have negligible internal resistance, the value of the resistor R will be

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A) 200 Ω B)100 Ω C)500 Ω D)1000 Ω

17. Equation of a particle moving along the x axis is $x = u(t-2) + \alpha(t-2)^2$. Here t is time in seconds.

a) The initial velocity of the particle is u

b) The acceleration of the particle is α

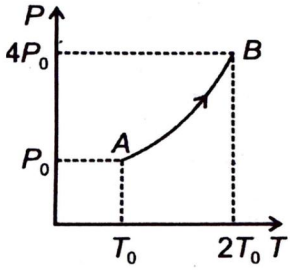
c) The acceleration of the particle is 2α

d) At $t = 2$ particle is at origin.

Choose the correct statements.

A) a, b, c, d B) a, b C) c, d D) a, b, d

18. One mole of an ideal monoatomic gas is taken through a process $A \rightarrow B$ along the curve $\frac{P}{T^2} = \text{constant}$ as shown in the graph. Heat absorbed by the gas in the process is (R is the universal gas constant)



A) RT_0 B) $\frac{RT_0}{2}$ C) $\frac{3RT_0}{2}$ D) $\frac{3RT_0}{4}$

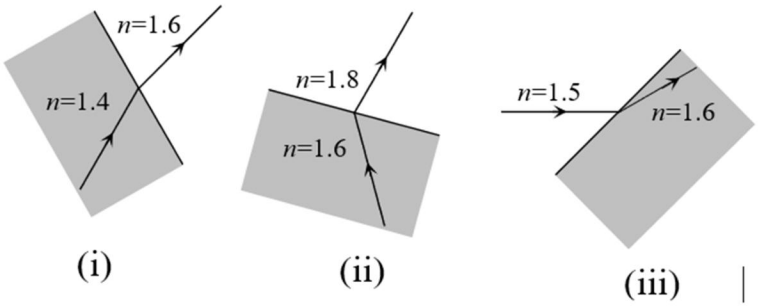
19. Consider a hydrogen like atom of atomic number Z obeying the Bohr's model. It is given that the difference between the $(n+1)^{th}$ Bohr radius and n^{th} Bohr radius is equal to the $(n-1)^{th}$ Bohr radius. The energy of the nth state (in eV) is:

A) $-0.85Z^2$ B) $-1.5Z^2$ C) $-13.6Z^2$ D) $-3.4Z^2$

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20. Which of the following ray diagram show physically possible refraction

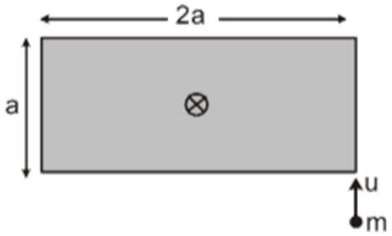


A) (i) B) (ii) C) (iii) D) None of these

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A uniform rectangular plate of mass m, hinged about its centre, is lying on a smooth horizontal surface. A particle of mass m and moving with a speed u collides with the plate as shown and sticks to it. The angular velocity of the plate just after collision is $\frac{xu}{5a}$. Find x.



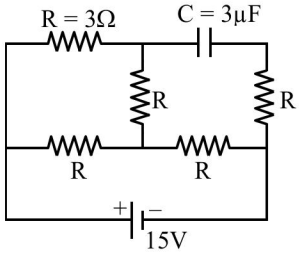
22. A uniform magnetic field of magnitude 0.20 T exists in space from east to west. With what speed should a particle of mass 0.010 g and having a charge 1.0×10^{-5} C be projected from south to north so that it moves with a uniform velocity (in m/s) is (Given g=9.8 ms^{-2})

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23. A particle executes simple harmonic motion between x = – A and x = + A. The time taken for it to go from 0 to A/2 is T_1 and to go from A/2 to A is T_2. If $T_2=nT_1$, the value of n is

24. In the circuit shown, the cell is ideal, with e.m.f. = 15V. Each resistance is of 3 Ω. The potential difference across the capacitor in the steady state is (in volt)-



25. A closed organ pipe and an open pipe of same length produce 4 beats when they are set into vibrations simultaneously. If the length of each of them were twice their initial lengths, the number of beats produced will be _____
(Assume same mode of vibration in both cases)

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The IUPAC name of the coordination compound $\left[Cr\left(H_2O\right)_6\right]\left[CoCl_6\right]$ is

A) Hexa chlorido cobalt (III) Hexa aqua chromate (III)

B) Hexa aqua chromate (III) Hexa chlorido cobalt (III)

C) Hexa aqua chromium (III) Hexa chlorido cobaltate (III)

D) Hexa chlorido cobaltate (III) Hexa aqua chromium (III)

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27. For a first order reaction $A \rightarrow B$, time taken for 10% completion of reactions is 20min. The time taken for 19% completion of the reaction is _____ min. (given $\log 10 = 1; \log 9 = 0.954$)

A) 20 min B) 40 min C) 60 min D) 120 min

28. A gaseous mixture contains hydrogen atoms in the 4th excited state, He^{+} ions in 3rd excited state and Li^{2+} ions in 2nd excited state. The number of spectral lines obtained in the emission spectrum of this sample when all these atoms/ions return to the ground state is

A) 19 B) 20 C) 16 D) 18

29. The precipitate of $CaF_2\left(K_{sp} = 1.7 \times 10^{-10}\right)$ is obtained when equal volumes of each of the following is mixed at constant temperature.

A) $1 \times 10^{-2} M\ Ca^{2+}$ and $1 \times 10^{-3} MF^{-}$ B) $1 \times 10^{-4} M\ Ca^{2+}$ and $1 \times 10^{-4} M\ F^{-}$

C) $1 \times 10^{-2} M\ Ca^{2+}$ and $1 \times 10^{-5} MF^{-}$ D) $1 \times 10^{-3} M\ Ca^{2+}$ and $1 \times 10^{-5} M\ F^{-}$

30. Observe the following statements:

a. The almost identical radii of Zr and Hf is consequence of Lanthanide contraction

b. Actinium is also a member of actinides

c. Copper (I) compounds are colourless except where colour results from charge transfer phenomenon

d. Eu^{2+} acts as a strong reducing agent

The correct statement(s) is(are):

A) only b B) only a, c and d C) only a and d D) only a and b

31. Which one of the following statements is incorrect?

A) Highest oxidation state of manganese in fluorides is $+4\left(MnF_4\right)$ but highest oxidation state in oxides is $+7\left(Mn_2O_7\right)$

B) $VO_2^{+} < Cr_2O_7^{2-} < MnO_4^{-}$ is the correct order of oxidising power

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C) When I^- is oxidized by MnO_4^- in acidic medium, I^- converts to I_2 and MnO_4^- converts to MnO_2

D) Silver nitrate on heating decomposes to give two types of paramagnetic gases along with a residue

32. Which of the following statement is incorrect?

A) Ionization enthalpy of an atom in gaseous state is equal in magnitude but opposite in sign to the electron gain enthalpy of the unipositive ion of the same atom in gaseous state.

B) F has the highest Electronegativity out of all the elements in the long form of periodic table.

C) In any period, the atomic radius of the noble gas is lowest.

D) The second ionization energy of Sulphur $\left(S_{(g)}\right)$ is greater than that of chlorine $\left(Cl_{(g)}\right)$.

33. Given below are two statements:

Statement I : Boron has a smaller first ionization enthalpy than beryllium.

Statement II : The penetration of a 2s electron to the nucleus is more than the 2p electron hence 2p electron is more shielded by the inner core of electrons than the 2s electrons.

In the light of the above statements, choose the correct answer from the options given below:

A) Both Statement – I and Statement – II are false

B) Statement – I is true but Statement – II is false

C) Both Statement – I and Statement – II are true

D) Statement I is false but Statement – II is true

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34. Consider following solution:

(I) 1M glucose (aq) (II) 1M sodium chloride (aq)

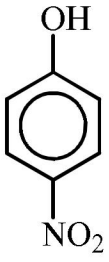
(III) 1M acetic acid in benzene (IV) 1M ammonium phosphate (aq)

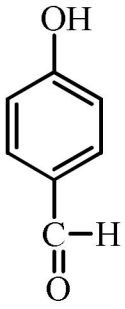
The **INCORRECT** among the following option is

A) All are isotonic solution B) III is hypotonic of I, II, IV

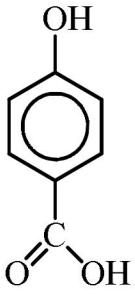
C) I, II, IV are hypertonic of III D) IV is hypertonic of I, II, III

35. The $H^{\oplus}$ ion is most readly lost by

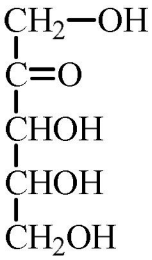
A) 

B) 

C) 

D) 

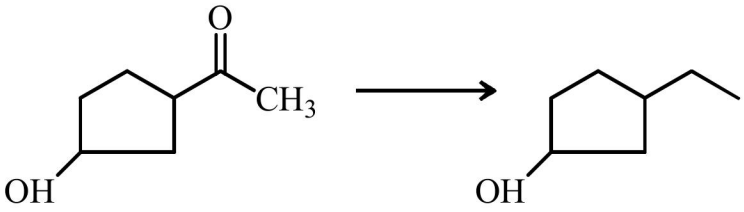
36.



The total number of possible optical isomers for above structure is

A) 12 B) 4 C) 16 D) 8

37. The appropriate reagent for the transformation:



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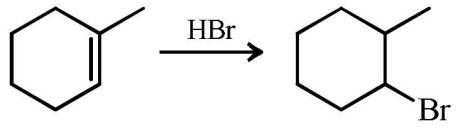
A) $Zn(Hg)/conc.HCl$

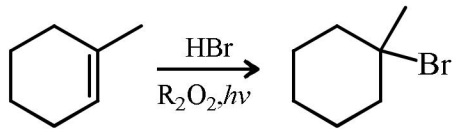
B) H_2/Ni

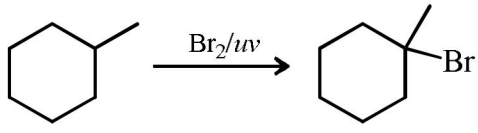
C) (i) NH_2NH_2 (ii) KOH / Ethylene glycol, Heat

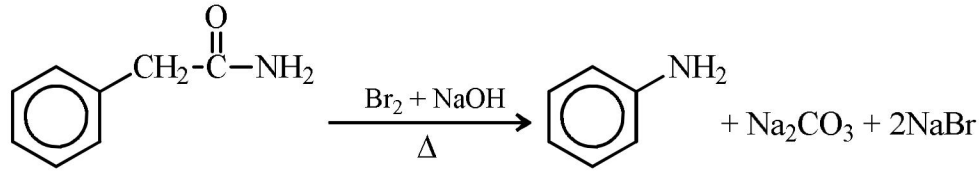
D) $NaBH_4$

38. Which of the following reaction is correct

A) 

B) 

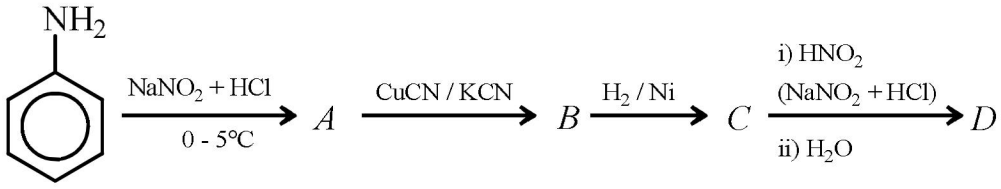
C) 

D) 

39. The dipolemoment of HBr is $2.60\times10^{-30}C.m$ and the interatomic spacing is $1.41\overset{0}{A}$. What is the percentage ionic character of HBr?

A) 11.5% B)21.2% C)5.2% D)27%

40. Aniline in a set of reaction yields a product D. The structure of the product D would be.



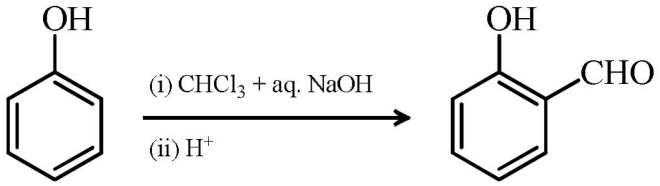
A) $C_6H_5CH_2NH_2$ B) $C_6H_5NHCH_2CH_3$

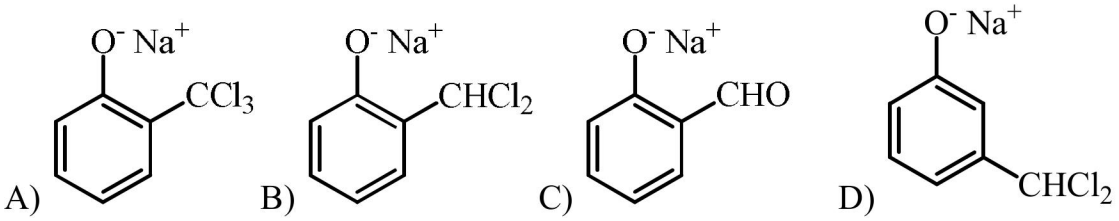
C) C_6H_5NHOH D) $C_6H_5CH_2OH$

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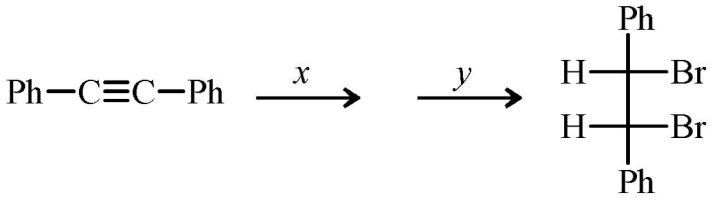
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41. The intermediate product in the reaction is.





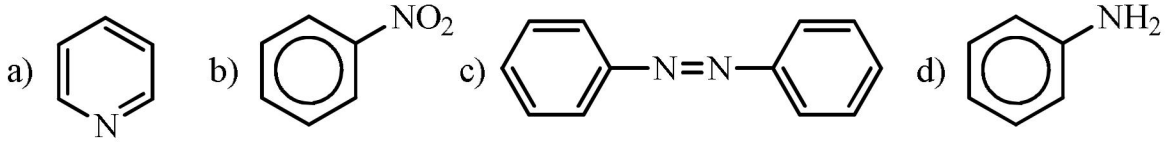
42. The reagents x and y in the following reaction are respectively:



A) $Br_2 / CCl_4, H_2 / Ni$ B) HBr, HBr

C) $Na / NH_3(l), Br_2 / CCl_4$ D) Br_2 / CCl_4, lindlar's hydrogenation

43. Kjeldalhl's method of estimation of N is not applicable to the compounds



A) All a, b, c, d B) only a, b, c C) only c and d D) only a, c, d

44. $\underset{\text{(Major)}}{(Y)} \xleftarrow{KCN} CH_3CH_2 - Cl \xrightarrow{AgCN} \underset{\text{(Major)}}{(X)}$, here (X) and (Y) respectively are:

A) CH_3CH_2CN and CH_3CH_2NC B) CH_3CH_2CN and CH_3CH_2CN

C) CH_3CH_2NC and CH_3CH_2CN D) CH_3CH_2NC and CH_3CH_2NC

45. The use of H_2S as a reagent for qualitative analysis of cations is based on the fact that

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A) H_2S forms coloured sulphides

B) H_2S is a weak acid and we can change degree of ionization of H_2S by adding HCl or NH_4OH

C) H_2S is a weak acid

D) The colour of the metal sulphide precipitated depends upon the pH of the solution.

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. How many ml of 0.1 M solution of $Mg(MnO_4)_2$ will be needed to oxidise completely 20 ml, 0.2 M ferrous oxalate in neutral medium.

47. Ammonia gas at a pressure of 20 atm and 27^oC is heated in a constant volume in a container to a temperature of 327^oC at which new pressure becomes 50 atm and the following equilibrium $2NH_{3(g)} \rightleftharpoons N_{2(g)} + 3H_{2(g)}$ is established. Percentage dissociation of ammonia at this temperature is X^2. The value of X is

48. 3.92g of Mohrs's salt react completely with $50ml\frac{N}{10}KMnO_4$ solution in acidic medium. The percentage purity of the sample is '$x\%$' then x is
(Gram atomic weight of $Fe = 56g$, $N = 14g$, $H = 1g$, $S = 32g$, $O = 16g$)

49. The heat of dissociation of $H_{2(g)}$ is 435 kJ mol^{-1}. If $C_{(graphite)} \rightarrow C_{(gas)}$ $\Delta H = 720\,\text{kJ mol}^{-1}$ and $\Delta_f^0 H$ for $CH_{4(g)} = -75$ kJ mol^{-1}. Then $\Delta_a^0 H$ of $CH_{4(g)}$ is $x\ kJ/mol$, then the value of $\frac{x}{4}$ is

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50. A 4.0 molar aqueous solution of $NaCl$ is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes. If the cathode is a Hg electrode, the maximum weight (g) of amalgam (Na – Hg) formed from this solution is : (atomic mass: Na=23, Hg=200)

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. $\lim\limits_{n\to\infty}(-1)^{n+1}\sin\left(\pi\sqrt{n^2+\frac{n}{2}+1}\right);n\in N$, is equal to

A) 1 B) $\frac{1}{\sqrt{2}}$ C) $\frac{-1}{\sqrt{2}}$ D) –1

52. $f(x)=\begin{cases}4x-x^3+\ln\left(a^2-3a+3\right), & 0\le x<3\\ x-18, & x\ge 3\end{cases}$

Complete set of values of $'a'$ such that $f(x)$ has a local minima at $x=3$ is

A) $[-1,2]$ B) $(-\infty,1]\cup[2,\infty)$

C) $[1,2]$ D) $(-\infty,1)\cup(2,\infty)$

53. If the sum of all rational terms in expansion of $\left(2^{\frac{1}{5}}+5^{\frac{1}{3}}\right)^{15}$ is equal to k, then the sum of all digits of k is _____

A) 10 B) 12 C) 13 D) 11

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54. The value of $\int \frac{\left(\sin^{3/2}\theta+\cos^{3/2}\theta\right)d\theta}{\sqrt{\sin^3\theta.\cos^3\theta.\sin(\theta+\alpha)}}$ is

A) $\frac{2}{\cos\alpha}\sqrt{\cos\alpha.\tan\theta+\sin\alpha}-\frac{2}{\sin\alpha}\sqrt{\cos\alpha+\cot\theta.\cos\alpha}+c$

B) $\frac{2}{\cos\alpha}\sqrt{\cos\alpha.\tan\theta+\sin\alpha}-\frac{2}{\sin\alpha}\sqrt{\cos\alpha+\cot\theta.\sin\alpha}+c$

C) $\frac{2}{\cos\alpha}\sqrt{\sin\alpha.\tan\theta+\cos\alpha}-\frac{2}{\sin\alpha}\sqrt{\cos\alpha+\cot\theta.\cos\alpha}+c$

D) $\frac{2}{\cos\alpha}\sqrt{\cos\alpha.\tan\theta+\cos\alpha}-\frac{2}{\sin\alpha}\sqrt{\cos\alpha+\cot\theta.\cos\alpha}+c$

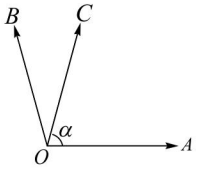
55. The distance of the point $(1,2)$ from the line $x+y+5=0$ measured along the line parallel to $3x-y=7$ is equal to:

A) $4\sqrt{10}$ B) 40 C) $\sqrt{40}$ D) $10\sqrt{2}$

56. Let the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{7}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{\alpha}=\frac{1}{25}$ coincide. Then the length of latus rectum of the hyperbola is:

A) $\frac{32}{9}$ B) $\frac{18}{5}$ C) $\frac{27}{4}$ D) $\frac{27}{10}$

57. Vectors $\overrightarrow{OA},\overrightarrow{OB},\overrightarrow{OC}$ are on the same plane, $\left|\overrightarrow{OA}\right|=1,\left|\overrightarrow{OB}\right|=1,\left|\overrightarrow{OC}\right|=\sqrt{2}$, the relative position is shown in the figure.



If $\tan\alpha=7$ and $\left(\overrightarrow{OB},\overrightarrow{OC}\right)=\frac{\pi}{4}$ and $\overrightarrow{OC}=m\overrightarrow{OA}+n\overrightarrow{OB},(m,n\in R)$, find the value of $(m+n)$.

A) 7 B) 9 C) 8 D) 3

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58. Consider the function $f(x)=\begin{cases}\dfrac{P(x)}{\sin(x-2)}, & x\neq 2\\ 7, & x=2\end{cases}$, where $P(x)$ is a polynomial such that $P''(x)$ is always a constant and $P(3)=9$. If $f(x)$ is continuous at $x=2$, then $P(4)$ is equal to

A) 20 B) 22 C) 24 D) 25

59. Let y be an implicit function of x defined by $x^{2x}-2x^{x}\cot y-1=0$. The value of $y'(1)$, where y' denotes the first derivative of y, is $\left(y\in(0,\pi)\right)$:

A) $-\ln 2$ B) $\ln 2$ C) -1 D) -2

60. The solution of differential equation $\dfrac{dy}{dx}=\dfrac{x^2+y^2+1}{2xy}$ satisfying $y(1)=0$ is:

A) a circle B) $y^2=x^2+x-10$

C) a hyperbola D) a ellipse

61. In a paper of English there are 5 questions such that the sum of marks is 30 and the marks for any question is not less than 2 and not more than 8. If the number of ways in which marks can be awarded is a 3 digit number xyz then the value of $\dfrac{2}{5}(x+y+z)$ is equal to? (Given that marks can be allotted in integers only)

A) 5.4 B) 6.4 C) 7.4 D) 8.4

62. Let a,b,c (in order) be the first three terms of a sequence and satisfying $\log\left(\dfrac{8b^3-a^3-c^3}{6abc}\right)=0$ and $\log b=\log(a^2-4)=\log(c-2)$. If T_n and S_n denote n^{th} term and sum of first n terms of the sequence , then wrong statement will be

A) $T_{10}=31$ B) $S_{10}=120$

C) $T_{21}^2-2T_{21}T_1+2T_1^2=1609$ D) $S_{11}-S_{10}=23$

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63. An equilateral triangle ΔOAB has side length 1, P is a point on the plane of the triangle. If $\overrightarrow{OP}=(2-t)\overrightarrow{OA}+t\overrightarrow{OB}, t\in R$, then the minimum value of $\left|\overrightarrow{AP}\right|$ is.

A) $\frac{1}{2}$ B) $\frac{1}{\sqrt{2}}$ C) $\frac{\sqrt{3}}{2}$ D) 2

64. If $f(x)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)-2\tan^{-1}x$ and $g(x)=\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)+4\tan^{-1}x$ then range of $(f(x)-g(x))$ for $x\in(-\infty,-1]$, is :

A) $\left[0,\frac{3\pi}{2}\right)$ B) $\left[\frac{-3\pi}{2},\pi\right)$ C) $\left[-\pi,\frac{-\pi}{2}\right)$ D) $\left[\pi,\frac{7\pi}{2}\right)$

65. If $\sum_{i=1}^{4}(x_i^2+y_i^2)\le 2x_1x_3+2x_2x_4+2y_2y_3+2y_1y_4$, the points $(x_1,y_1),(x_2,y_2),(x_3,y_3),(x_4,y_4)$ are

A) the vertices of a rectangle

B) collinear

C) the vertices of a trapezium but not a rectangle

D) none of these

66. If $R=\{(x,y):x,y\in Z,x^2+3y^2\le 8\}$ is a relation on the set of integers Z, then the domain of R^{-1} is

A) $\{-2,-1,1,2\}$ B) $\{0,1\}$ C) $\{-2,-1,0,1,2\}$ D) $\{-1,0,1\}$

67. If $\sum_{i=1}^{9}(x_i-5)=9$ and $\sum_{i=1}^{9}(x_i-5)^2=45$ then the standard deviation of the 9 items $x_1,x_2,.....x_9$ is

A) 9 B) 4 C) 2 D) 3

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68. $\int_0^{10}[x]^3\{x\}\,dx$ is equal to (where $[.],\{\}$ denotes greatest integer and fractional function respectively).

A) 2025 B) $\frac{2025}{2}$ C) $\frac{2025}{4}$ D) $\frac{2025}{8}$

69. Let $z=1+\cos\frac{8\pi}{5}+i\sin\frac{8\pi}{5}$ then the modulus and argument of 'z' will be

A) $|z|=2\cos\frac{\pi}{5}$ *and* $\arg(z)=\frac{-\pi}{5}$ B) $|z|=2\cos\left(\frac{4\pi}{5}\right)$ *and* $\arg(z)=\frac{4\pi}{5}$

C) $|z|=2\cos\left(\frac{2\pi}{5}\right)$ *and* $\arg(z)=\frac{-\pi}{5}$ D) $|z|=2\cos\left(\frac{2\pi}{5}\right)$ *and* $\arg(z)=\frac{4\pi}{5}$

70. Given set $=\{1,2,3,4,5,.....1000\}$

One number is selected randomly from the set. Then the probability that selected number is divisible by at least one of 2,3 or 5.

A) $\frac{367}{500}$ B) $\frac{257}{500}$ C) $\frac{417}{500}$ D) $\frac{217}{500}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let OA,OB,OC be coterminous edges of a cuboid. If ℓ,m,n be the shortest distances between the sides OA,OB,OC and their respective skew body diagonals to them, respectively , then value of $\frac{\left(\frac{1}{\ell^2}+\frac{1}{m^2}+\frac{1}{n^2}\right)}{\left(\frac{1}{OA^2}+\frac{1}{OB^2}+\frac{1}{OC^2}\right)}$ is.

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72. Let K be a positive real number and

$$A=\begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix} \text{ and } B=\begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$$

If $det(adj\ A)+det(adj\ B)\ =\ 10^6$, then $[k]$ is equal to_____.

[Note : $(adj\ M)$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k].

73. If the area of the region $\left\{(x,y)\in R^2 : y^2 \geq 4x, |y| \leq \frac{x}{2}+2\right\}$ is Δ, then the value of $[\Delta]$.

[Note : $[x]$ denotes greatest integer less than or equal to x.]

74. A line is drawn from a point $P(x,y)$ on curve $y=f(x)$ making an angle with positive direction of the x-axis which is supplementary to the one made by the tangent to the curve at $P(x,y)$. The line meets the x-axis at A. Another line perpendicular to the first, is drawn from $P(x,y)$ meeting the y-axis at B. If $OA=OB$, where O is the origin and the curve which passes through $(1,1)$ is $x^2-y^2+2xy=c$ then 'c' will be_____.

75. Let (x,y) satisfy the curve $x^2+y^2-6x-8y+21=0$. Let m_1 and m_2 are minimum and maximum values of $\left(\frac{x}{y}\right)$. If $m_1^2+m_2^2=\frac{p}{q}$ where $p,q\in N$, then the least value $(p+q)$ is.

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